



Trinity College Dublin
Coláiste na Tríonóide, Baile Átha Cliath
The University of Dublin

Instructions for Vice Assembly

3B7 Manufacturing Systems and Technology

DFMA Assignment

Group 30

The following document supplements our report, serving as an instruction manual for the assembly of our chosen product. We chose a workbench vice. This document contains written instructions, a link to video instructions, images of the original vice with annotated dimensions, images of the CAD model before changes were made, images of the CAD model after changes were made, and also engineering drawings.

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Written Detailed Assembly Instructions

- 1) For the assembly the following components should be available:
 - a. 1 x fixed jaw component
 - b. 1 x dynamic jaw component
 - c. 1 x polished handle
 - d. 1 x greased lead screw
 - e. 1 x spring of high stiffness
 - f. 1 x spring pin
 - g. 1 x retaining pin
 - h. 2 x serrated jaw attachments
 - i. 4 x small Phillips head screws (for the serrated jaws)
 - j. 1 x large hex screw (for retaining pin)
 - k. 4 x round washers
 - l. 2 x large hex socket-head screws
 - m. 2 x hex nuts
- 2) Take the fixed jaw and retaining pin. Using the large black hex screw, screw the retaining pin to the internal body of the fixed jaw.
- 3) Take the fixed jaw and dynamic jaw. Using the 4 small Phillips-head screws, attach one serrated jaw to the both the dynamic and fixed jaw.
- 4) Take the lead screw and the dynamic jaw part. Pass the screw through the hole in the front of the part. Put the spring on the screw and use the pin to load it and secure it in place.
- 5) Through the hole in the extruding end of the lead screw, insert the handle. Take two thin, narrow pieces of similar metal and weld them to either end of the rod to ensure that the handle can no longer be removed from the lead screw.
- 6) Take 2 washers and place them on both of the large hex socket-head screws. Take the 2 screws and insert them through the base of the fixed jaw and the table. Then take 2 washers and place them over the thread of either screw. Tightly screw on the 2 hex nuts onto either screw.
- 7) Take the dynamic jaw part and insert it into the fixed jaw. Spin the handle so that the lead screw inserts into the retaining pin.

Assembly Instruction Video

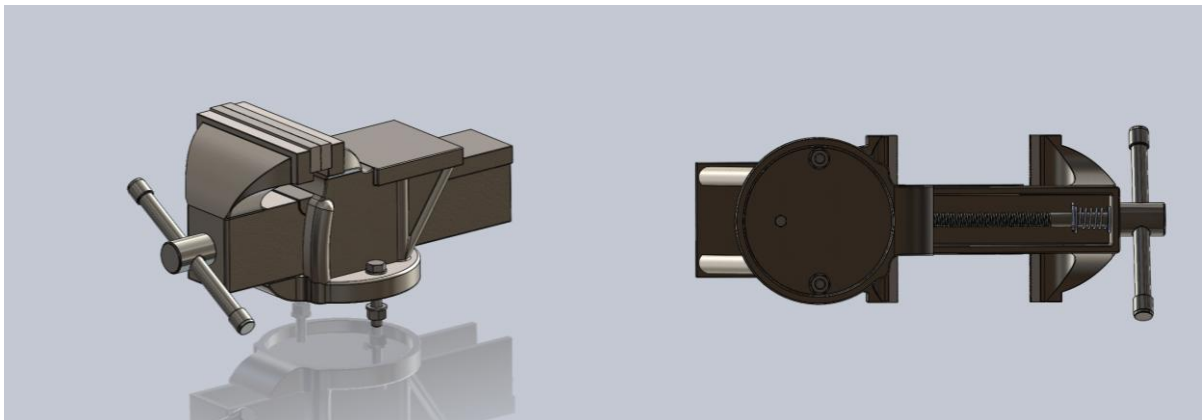
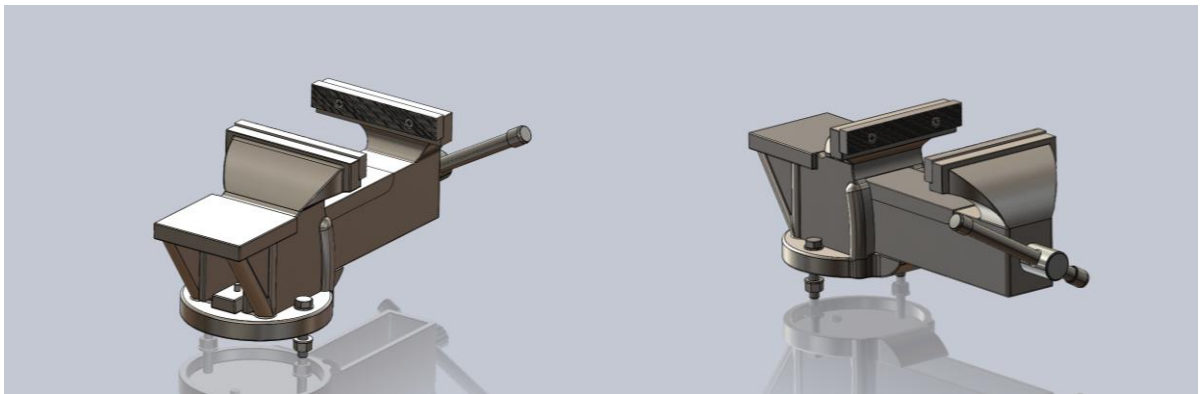
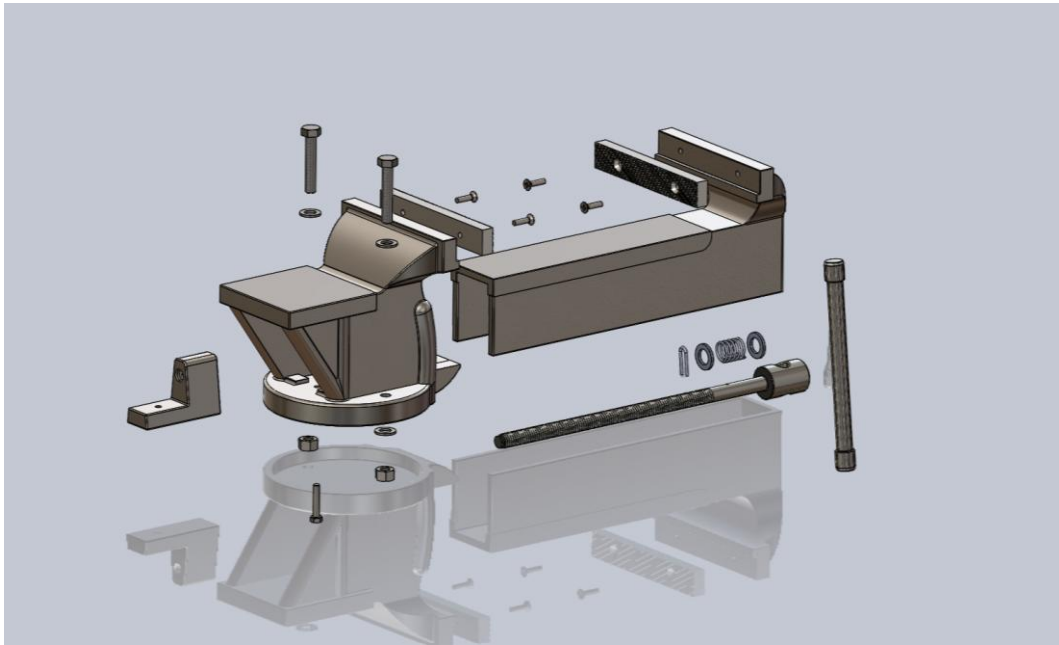
For a complete visual demonstration of how to assemble the vice, please see our assembly instruction video which can be accessed through the following link:

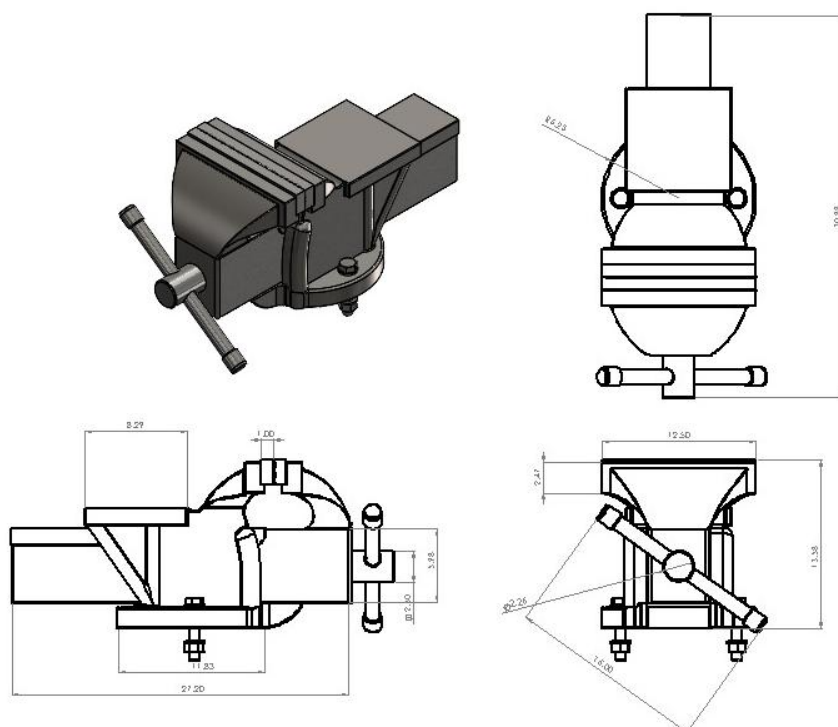
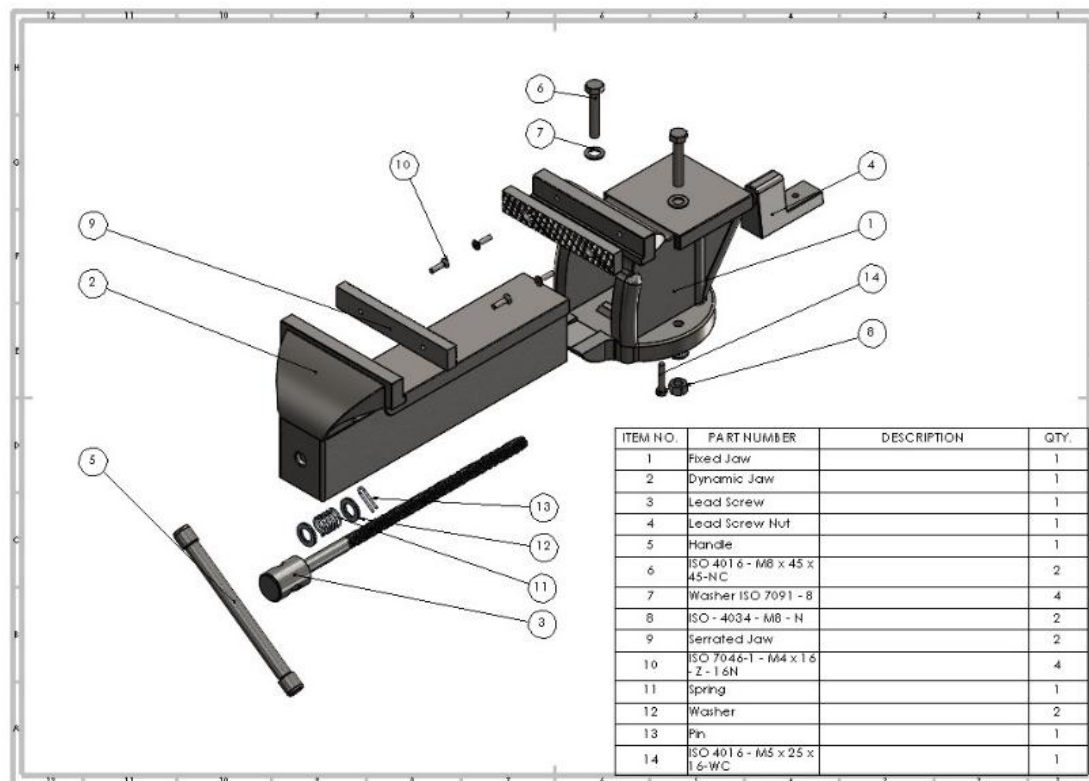
- [3B7 Group 30 Assembly Instructions Video](#)

The video gives a more detailed approach to our reasoning behind certain assembly methods, and allows the end user to gain an intuitive understanding through a visual demonstration

Appendix

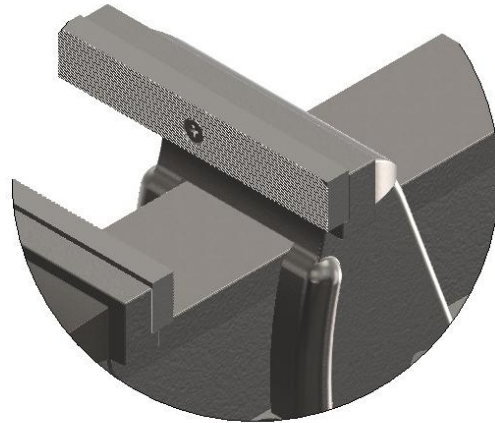
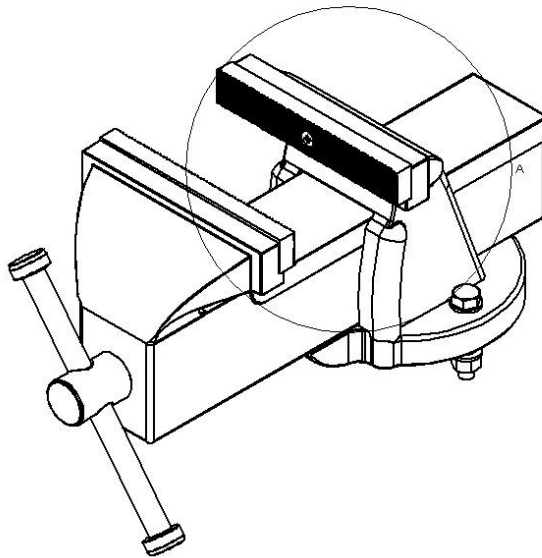
Images and Drawings of the Original CAD Model



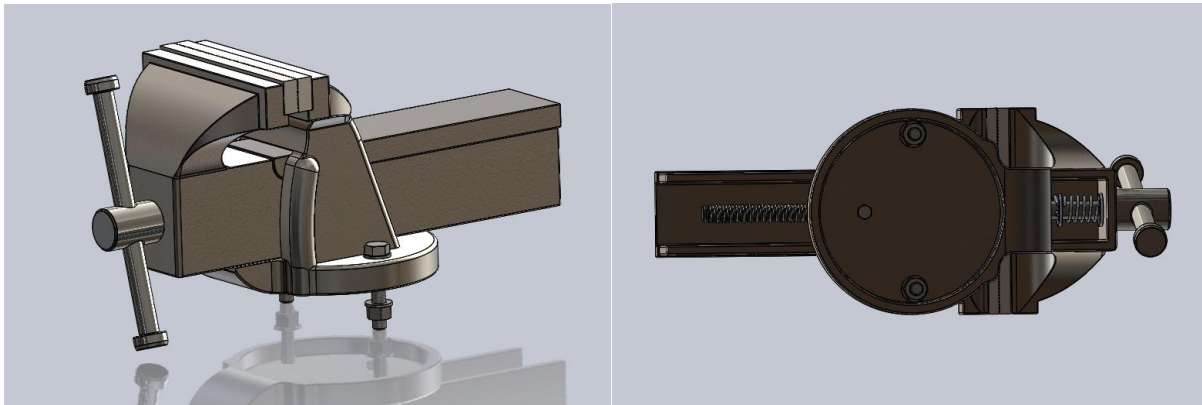
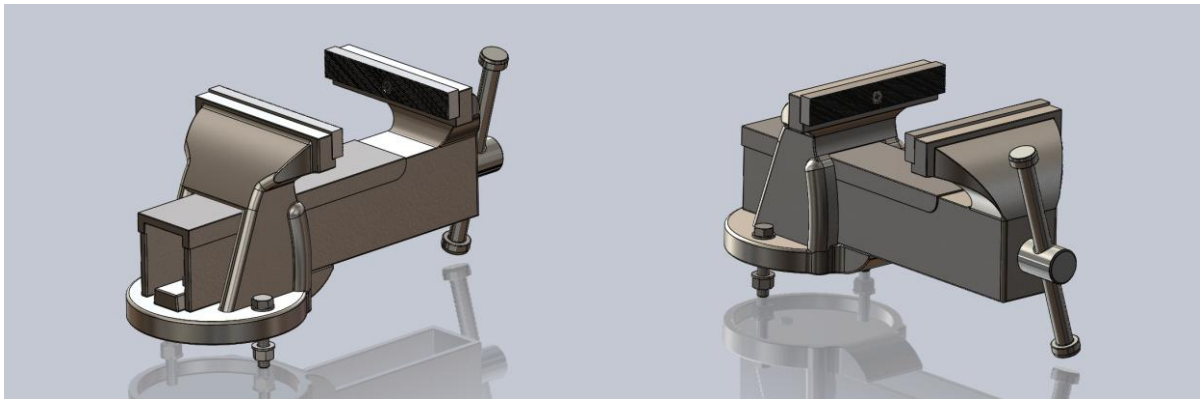


Images and Drawings of the Redesigned CAD Model

Redesigned serrated jaw with a singular screw rather than two.

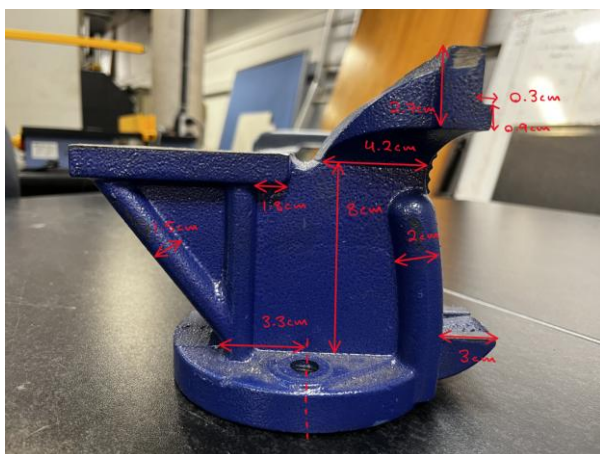
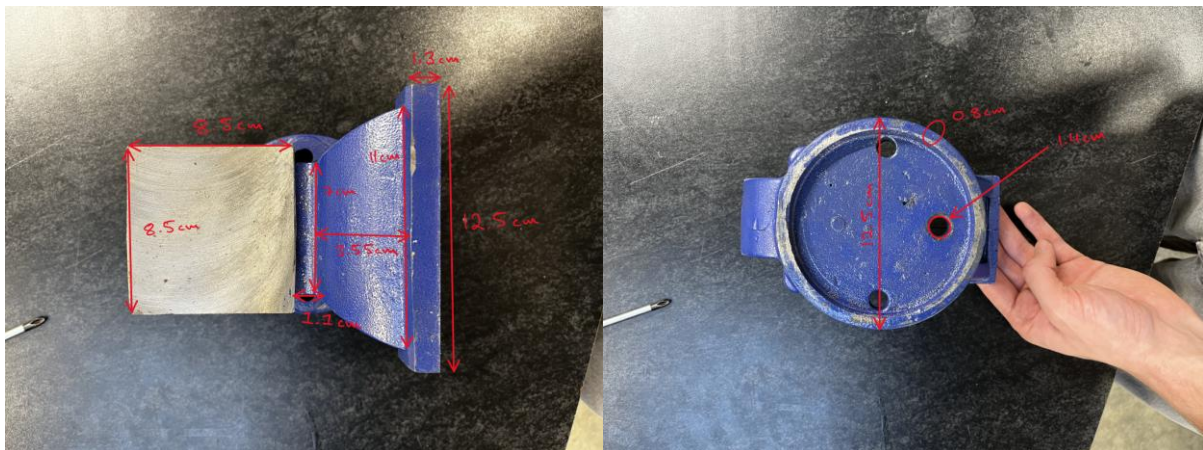
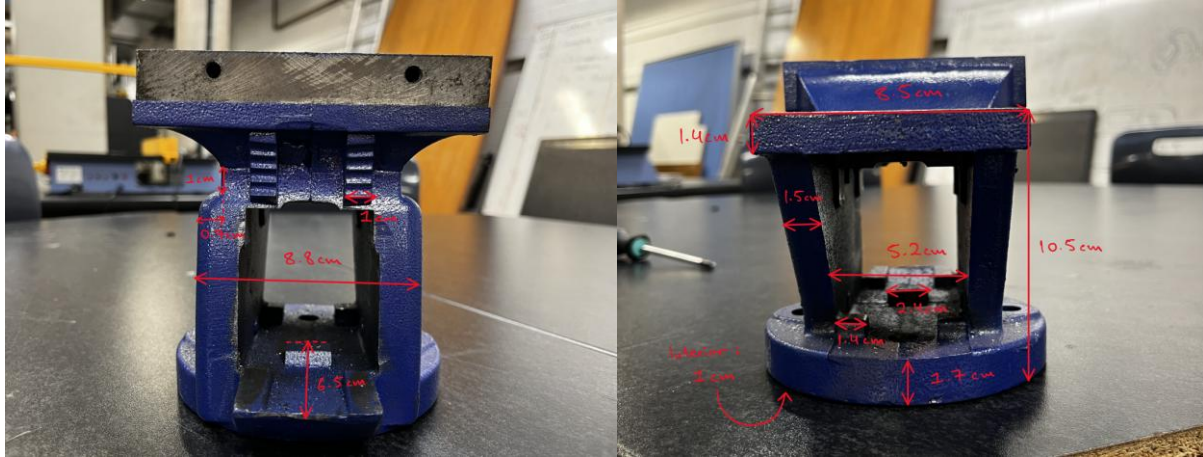


DETAIL A
SCALE 1 : 1



Annotated Images of the Original Vice

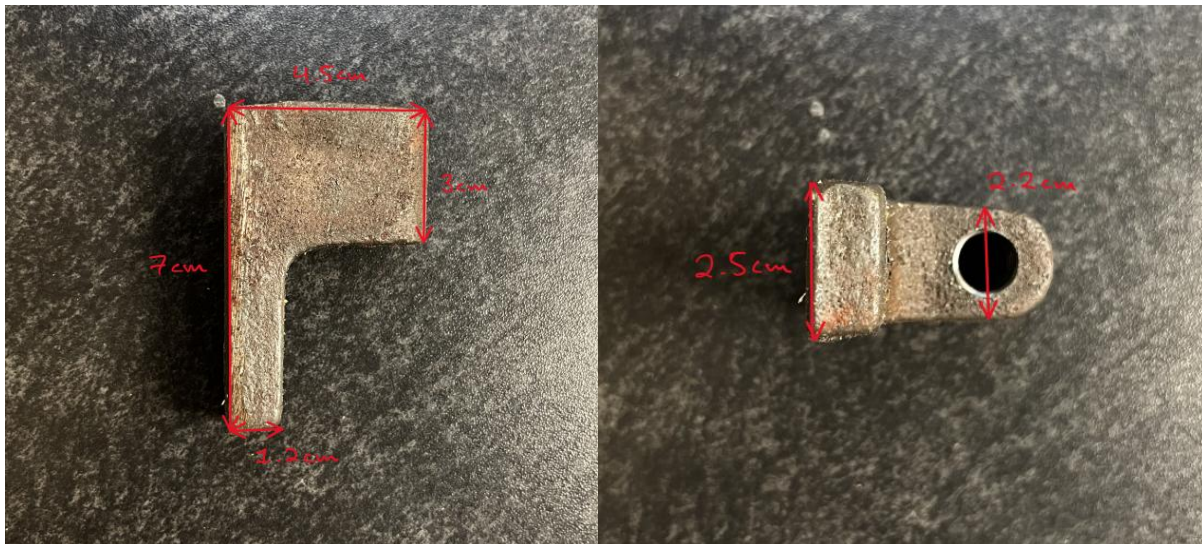
We took photos of the vice and added dimensions onto them to aid with the CAD design process. This proved to be very helpful, as the CAD replica was almost a 1-1 copy of the actual vice.



Serrated Jaws:



Retaining Pin:



Dynamic Jaw and Handle:

